Mohammad Y. Abdelshafy

Rochester, New York, United States
yasser.abdelshafy@rochester.edu | Energy Research Digest

Education

University of Rochester, Rochester, NY
Doctor of Philosophy (PhD), Mechanical Engineering 2027 (expected)
Master of Science (MS), Mechanical Engineering 2023
Helwan University, Cairo, Egypt
Bachelor of Science (BS), Mechanical & Energy Engineering 2022
STEM High School for Boys - 6th of October, Egypt
High School Diploma, STEM 2017

Research Interests

Fluid dynamics, Rayleigh-Benard convection, liquid metal flows, magnetohydrodynamics (MHD), geophysical fluid dynamics, planetary dynamos.

Research Experience

Doctoral Researcher, University of Rochester, Rochester, NY Aug 2022 – Present

- Currently exploring the role of rotating convection in the Early Earth's dynamo and its magnetic field.
- Designing and building a rotating experiment to explore the effect of boundary roughness on convective heat transfer and flow stability in a rotating Rayleigh-Bénard convection system in both water and liquid gallium.
- Invested fluid dynamics in liquid metal batteries to improve operational safety and stability.
- Analyzed ultrasound data from liquid metal experiments featuring both current-driven electrovortex flow and convection.

Teaching Experience

Graduate Teaching Assistant (GTA) Coach, University of Rochester Jun 2024 – Sep 2024

- Developed and delivered a teaching workshop for new GTAs in mechanical engineering.
- Attended a "train-the-trainer" course to prepare session materials aligned with departmental GTA duties.

Graduate Teaching Assistant, University of Rochester Jan 2023 – May 2024

- ME241: Fluids Lab (Spring 2023, Spring 2024): Guided students in aerodynamic tests, wind tunnel operation, and analysis of lift and drag. Followed up weekly with student projects.
- ME120: Statics (Fall 2022): Delivered revision sessions, held office hours and tutoring sessions, graded assignments, quizzes, and exams, and created exam review material for first-year students.
- ENE336: Power Stations Technology: Graded homework and assisted students with problem sets.

Industry Experience

Engineering Intern, Siemens Energy, New Cairo, Egypt Aug 2021 – Sep 2021

- Completed intensive training sessions in high voltage power transmission, supplemented by site visits.
- Worked on the design of a step-down substation for an industrial complex in northern Egypt.

Manufacturing Engineer, NATPACK for plastic packaging solutions, El Obour, Egypt Jan 2021 – Jan 2021

- Supervised the setup and troubleshooting of new tube shouldering machines.

Maintenance Engineer (Intern), NATPACK, El Obour, Egypt Aug 2020 – Oct 2020

- Gained experience in plastic packaging production, mold design, and machinery maintenance.
- Operated ENGEL injection molding machines and troubleshooted operational halts.
- Supervised 26 maintenance operations in hydraulic and pneumatic systems.
- Managed factory utilities including HVAC, water cooling, and air compression systems.
- Implemented quality and safety measures in production and maintenance.
- Developed a production plan for a tube factory facing labor deficits, aiming to surpass production targets.

Energy Engineering Intern, Agiba Petroleum Co., Matruh, Egypt Jul 2019 – Jul 2019

- Gained hands-on experience with the operation and monitoring of 2MW gas turbines.
- Focused on turbine and pumping station maintenance.
- Toured drilling sites and natural gas treatment plants.

Publications and Presentations

Journal Publications

- M. Y. Abdelshafy, B. Wang, I. Mohammad, J. S. Cheng, D. H. Kelley. Current-driven flow transitions in laboratory liquid metal battery models. *J. Fluid Mech.* (2025), *vol.* 1007, A42. doi:10.1017/jfm.2025.78

Conference Presentations

- M. Y. Abdelshafy, J. S. Cheng, D. H. Kelley. Laboratory investigation of Topographical effects on outer core flow: Water Experiments. *AGU Fall Meeting*, 2024.
- J. S. Cheng, M. Y. Abdelshafy, D. H. Kelley. Laboratory investigation of topographical effects on outer core flow-liquid metal experiments. *AGU Fall Meeting*, 2024.
- M. Y. Abdelshafy, B. Wang, I. Mohammad, J. S. Cheng, D. H. Kelley. Transition in flows driven by converging currents. *APS Division of Fluid Dynamics Meeting*, 2024.
- J. S. Cheng, M. Y. Abdelshafy, B. Wang, I. Mohammad, D. H. Kelley. Interaction of convective modes with current-driven flow in a liquid metal battery experiment. *APS Division of Fluid Dynamics Meeting*, 2024.
- M. Y. Abdelshafy, J. S. Cheng, I. Mohammad, B. Wang, D. H. Kelley. Electrovortex flow in liquid gallium. *APS Division of Fluid Dynamics Meeting*, 2023.

Skills

Fluid Dynamics:	Experimental design, flow visualization, ultrasound diagnostics, rotating flows
Software:	MATLAB, Python, AutoCAD, Siemens NX, LaTeX
Fabrication:	Manual machining, molding and casting, experimental setup design and construction
Languages:	Arabic (Native), English (Fluent)

Leadership, Service, and Outreach

Board Member, M K Gandhi Institute, Rochester, NY Ongoing

- Serve on the board of directors for the M K Gandhi Institute for Nonviolence.
- Contribute to the development committee, focusing on connecting the institute with organizations involved in nonviolence, healing, and community engagement across the US.

President, International Students & Scholars Association (ISSA), University of Rochester May 2024 – Present

- Advocate for international students, and acting as a link between the students and the administration.
- Addressing student needs, facilitating smoother academic and personal transitions to the U.S., supporting with tax filing, etc.

Co-Founder, Larus Textile, Cairo, Egypt Jul 2019 – Nov 2019

- Co-founded an environmentally aware textile manufacturer.
- Participated in the SEYP19 incubation program by You ThinkGreen Egypt.
- Conducted market research, developed business plans, and analyzed financial statements.
- Successfully pitched the startup idea, securing a 4 million EGP fund.

Contributing Writer, Altanweeri Online Magazine, Amman, Jordan Apr 2018 – Sep 2018

Awards and Honors

Finalist, Climate Science Olympiad 2020
Venture Fund (4M EGP), SEYP Incubation Program, You ThinkGreen Egypt (for Larus Textile) 2019